Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 40)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
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9	
10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 . 1 Which of the following statements about stratospheric ozone is correct? Tick one box.

[1 mark]

A Ozone is destroyed only by CFCs ☐

B Ozone absorbs mainly infrared radiation ☐

C Ozone is formed when monatomic oxygen reacts with diatomic oxygen ☐

D Ozone concentrations are highest at ground level ☐

0 1 . 2 Explain the difference between the roles of ozone in the stratosphere and in the troposphere.

[2 marks]

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0 1 . 3 Explain why HCFCs cause less ozone depletion than CFCs.

[2 marks]

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0 2 Table 1 compares two types of satellite used to monitor the Earth.

Table 1

Feature	Low Earth Orbit (polar) satellite	Geostationary satellite
Altitude	About 800 km	36 000 km above the equator
Orbit period	About 1.5 hours	24 hours (fixed position)
Coverage	Whole Earth surveyed in about 15 days	Same third of the Earth continuously
Detail	High resolution	Lower resolution

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0 2 . 1 Use Table 1 to explain why LEO satellites are used to monitor changes in ice cover and vegetation, whereas geostationary satellites are used to monitor weather.

[3 marks]

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0 2 . 2 Give **three** climate-related factors, other than ice cover, that are measured by satellite sensors.

[3 marks]

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0 2 . 3 Explain how a computer climate model can be tested for reliability before its predictions are used.

[2 marks]

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0 2 . 4 Explain why representative data are essential for climate modelling.

[2 marks]

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0 3 Table 1 shows some techniques used to locate mineral deposits.

0 3 . 1 Complete Table 1 by adding the name of the technique or a description of how it works.

[4 marks]

Table 1

Technique	How it works
Gravimetry	
	Detects rocks that are more magnetic, such as magnetite iron ore or ores of tungsten and cobalt
Seismic survey	
	Measures how difficult it is for electricity to pass through rocks; sedimentary rocks with high water content have lower values than igneous rocks

0 3 . 2 Explain why trial drilling is usually carried out only after remote sensing surveys.

[3 marks]

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0 3 . 3 Explain how infrared spectroscopy from satellites can help identify mineral deposits.

[2 marks]

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0 3 . 4 Give **one** advantage of portable field analysis equipment such as X-ray fluorescence.

[1 mark]

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0 4 Table 1 shows the annual emissions of two pollutants in the UK in 1970 and 2015.

Table 1

Pollutant	Emissions in 1970 / thousand tonnes	Emissions in 2015 / thousand tonnes	Main source in 2015
Sulfur dioxide	6 400	220	Energy industries
Nitrogen oxides	2 700	930	Road transport

0 4 . 1 Calculate the percentage reduction in sulfur dioxide emissions between 1970 and 2015.

[2 marks]

Answer ..... %

0 4 . 2 Suggest **three** reasons why sulfur dioxide emissions fell by a much larger percentage than nitrogen oxide emissions.

[3 marks]

1 .....

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0 4 . 3 Explain why a reduction in NOx emissions would also reduce concentrations of tropospheric ozone and PANs.

[2 marks]

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0 4 . 4 Explain why lichens are used to monitor sulfur dioxide pollution.

[3 marks]

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0 5 . 1 Explain why phosphorus is more likely than nitrogen to be a limiting factor on biological productivity.

[3 marks]

0 5 . 2 Explain how mycorrhizal fungi increase phosphate uptake by plants.

[2 marks]

0 5 . 3 Describe how human activities have altered the phosphorus cycle.

[3 marks]

0 5 . 4 Explain **two** strategies for the sustainable management of phosphorus supplies.

[2 marks]

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0 6 Figure 1 shows the number of large oil spills (over 700 tonnes) from tankers worldwide and the amount of oil transported by sea between 1975 and 2015.

Figure 1

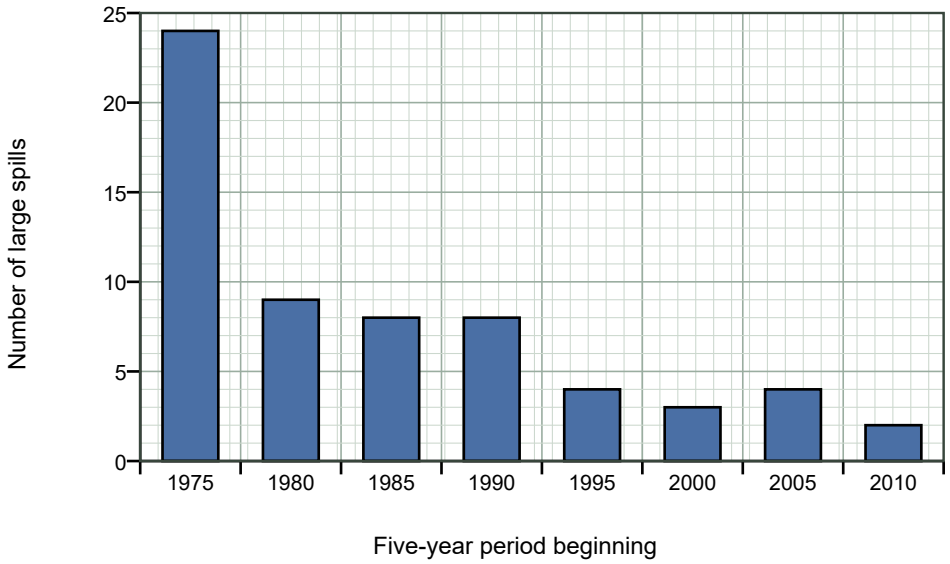


Table 1

Year	1975	1985	1995	2005	2015
Oil transported by sea / million tonnes	1 400	1 200	1 800	2 400	2 900

0 6 . 1 Calculate the percentage increase in oil transported by sea between 1975 and 2015.

[2 marks]

Answer ..... %

0 6 . 2 Describe the trend in the number of large oil spills shown in Figure 1.

[2 marks]

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0 6 . 3 Suggest **three** reasons for the decline in the number of large spills despite the increase in oil transported.

[3 marks]

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0 6 . 4 Explain why the total amount of oil pollution entering the sea may not have fallen as much as the number of large tanker spills.

[3 marks]

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0 7 . 1 Complete Table 1 to explain how each method reduces soil erosion.

[5 marks]

Table 1

Method	How it reduces erosion
Terracing	
Tied ridging	
Windbreaks / hedgerows	
Mulching	
Strip cropping	

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0 7 . 2 Explain why long-term crops such as tea, coffee or orchards reduce erosion compared with annual arable crops.

[2 marks]

0 7 . 3 Describe **three** ways livestock can be managed to reduce soil erosion.

[3 marks]

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0 8 . 1 For each of the following investigations, name the most appropriate statistical test.

[4 marks]

Whether noise level decreases as distance from a road increases .....

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Whether the numbers of dormice differ between woodlands with different management ..

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Whether mean dissolved oxygen differs between two streams .....

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Whether crop yield increases with fertiliser application (ranked data) .....

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0 8 . 2 Explain why the chi-squared test cannot be used for data such as wind speed or plant height.

[1 mark]

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0 8 . 3 A Spearman's rank test on 12 pairs of data gave r_s = -0.72. The critical value at p = 0.05 for n = 12 is 0.59. Explain what can be concluded.

[3 marks]

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0 8 . 4 Explain the purpose of error bars on a graph of mean values.

[2 marks]

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0 9 . 1 Define the term **persistence** as applied to a pollutant.

[1 mark]

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0 9 . 2 Explain how the persistence of a pesticide is affected by **two** named environmental factors.

[4 marks]

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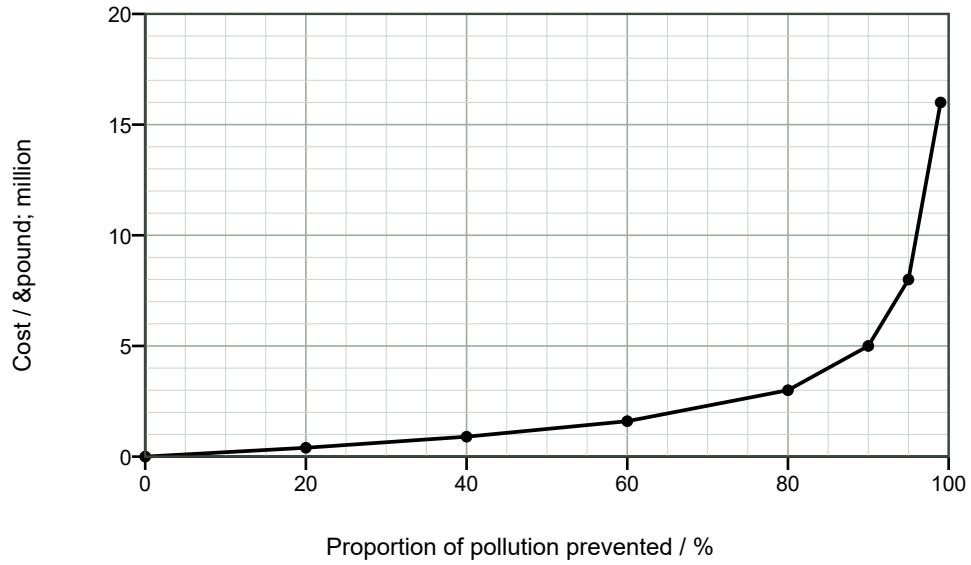
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1 0 Figure 1 shows how the cost of controlling emissions from a factory changes with the proportion of the pollutant that is prevented from being released.

Figure 1



1 0 . 1 Use Figure 1 to calculate the extra cost of increasing control from 90% to 99% of the pollutant.

[1 mark]

Answer £ million

1 0 . 2 Explain how the terms **ALARA** and **BATNEEC** are used when deciding how much pollution control is required.

[3 marks]

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1 0 . 3 Explain why the control methods that were originally adequate for a factory may no longer be sufficient.

[2 marks]

1 0 . 4 Evaluate the principles and general strategies that can be used to decide how pollution should be controlled.

[9 marks]

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Extra space

15

Write an essay on one of the following topics.

11 . 1 Discuss how the properties of solar power and hydroelectric power affect their use, and evaluate how new technologies could increase their contribution to energy supplies.

[25 marks]

OR

11 . 2 Discuss how an understanding of the hydrological cycle can be used to develop strategies for the sustainable management of water resources.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS